

Masaki Imagawa

Curriculum Vitae

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Contact Information

Institute of Mathematics, Academia Sinica
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Employment

Postdoctoral Fellow

Aug. 2025–Present

Institute of Mathematics, Academia Sinica, Taipei, Taiwan

Education

Ph.D. in Informatics, Kyoto University

Apr. 2022–Jul. 2025

Dissertation: Convergence analysis of elliptic regularization for advection equations in a bounded domain

Advisor: Hiroshi Fujiwara

Master of Informatics, Kyoto University

Apr. 2020–Mar. 2022

B.Sc. in Mathematics, Kyoto University

Apr. 2016–Mar. 2020

Publications

Journal Articles

Masaki Imagawa and Daisuke Kawagoe, A revisit on well-posedness of a boundary value problem of a stationary advection equation without the separation condition, *J. Math. Anal. Appl.* **551** (2025), no. 2, part 2, Art. 129695.

[doi:10.1016/j.jmaa.2025.129695](https://doi.org/10.1016/j.jmaa.2025.129695)

Conference Proceedings

Masaki Imagawa and Yuusuke Iso, Sharp estimates in singular perturbation applied to the first order partial differential equations in a bounded domain, in *2025 Days on Diffraction*, IEEE, 2025, 90–95.

[doi:10.1109/DD66835.2025.11263497](https://doi.org/10.1109/DD66835.2025.11263497)

Preprints

Masaki Imagawa and Daisuke Kawagoe,

On strong convergence of an elliptic regularization with the Neumann boundary condition applied to a stationary advection equation, [arXiv:2303.17904](https://arxiv.org/abs/2303.17904), 2023.

Presentations

Talks

Shock Waves and Boundary Layers for the Viscous Burgers Equation

Oct. 2026

National Chung Hsing University Applied Mathematics Seminar

Taichung, Taiwan

Stability and Boundary Interaction of Viscous Shock Waves for the Burgers Equation

Aug. 2026

19th EASIAM Annual Meeting

Jeju, Korea

Stability of Shock Waves for the Viscous Burgers Equation with Boundary Interactions

Apr. 2026

Taipei Postdoc Seminar, National Center for Theoretical Sciences

Taipei, Taiwan

Stability of Viscous Shock Waves for the Burgers Equation in a Half-Line Computational Mechanics Seminar, Kyoto University	Mar. 2026 Kyoto, Japan
Asymptotic stability of traveling waves for scalar viscous conservation laws with boundary Applied Mathematics Freshman Seminar 2025, Kyoto University	Nov. 2025 Kyoto, Japan
Convergence estimates for singular perturbation of advection equations with mixed boundary conditions Kyoto University Applied Mathematics Seminar	Jul. 2025 Kyoto, Japan
Convergence analysis of perturbed advection equations in a bounded domain Numerical Analysis Seminar, The University of Tokyo	Jul. 2025 Tokyo, Japan
Sharp estimates in singular perturbation applied to the first order partial differential equations in a bounded domain Days on Diffraction	Jun. 2025 Online
Boundary value problems of advection equations and its regularized problems Analysis Seminar, Academia Sinica	Mar. 2025 Online
On a Boundary Value Problem of a First-Order PDE and Its Regularization Problems 2024 Japan-Taiwan Joint Workshop on Numerical Analysis and Inverse Problems Kyoto University	Mar. 2024 Kyoto, Japan
Convergence Rates of Elliptic/Parabolic Regularization Applied to Advection Equations with Boundary Conditions The 5th Workshop on Recent Development of Mathematical Fluid Dynamics and Hyperbolic Conservation Laws	Sep. 2023 Jeju, Korea
On the Optimal Convergence Rate of Elliptic Regularization Methods Applied to Advection Equations The 19th Joint Meeting of JSIAM Activity Groups, Okayama University of Science	Mar. 2023 Okayama, Japan
On Convergence Rates of an Elliptic Regularization Applied to a Stationary Advection Equation The 13th Taiwan-Japan Joint Workshop for Young Scholars in Applied Mathematics National Center for Theoretical Sciences	Mar. 2023 Online
On Convergence Estimation of Elliptic Regularization Applied to Boundary Value Problem of Advection Equations 2022 Japan-Taiwan Joint Online Workshop on Numerical Analysis and Inverse Problem	Mar. 2022 Online
Numerical Computation of Advection Equations Using Elliptic Regularization Annual Meeting of the Japan Society for Industrial and Applied Mathematics	Sep. 2021 Online
Posters	
Convergence rates of elliptic regularization applied to advection equations with boundary conditions NCTS–Kyoto Mathematics Symposium, National Center for Theoretical Sciences	Des. 2023 Taipei, Taiwan
On the Convergence Rate of Singularly Perturbed Solutions to the Steady Advection Equation The 17th ICT Innovation, Kyoto University	Feb. 2023 Kyoto, Japan